Akoizo Original Practice - 2027 Rotation Form 1

Division C | Protein Modeling | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) Alpha-helix H bonds mainly involve…

Unit: Structural levels and denaturation.
Worked-case data: Heating disrupts activity but the amino-acid sequence remains intact. What changed?

A. Only neighboring side-chain colors

B. Backbone groups

C. Only sulfur atoms

D. Only DNA bases

Answer: _____

2. (2 points) Attachment and fusion are…

Unit: Viruses, entry, and immune recognition.
Worked-case data: An antibody binds a viral protein in an assay. Does that prove neutralization?

A. Only DNA processes

B. Absent in all enveloped viruses

C. Different entry-related steps

D. Always identical

Answer: _____

3. (2 points) Rotation changes…

Unit: Molecular viewing and reproducible JUDE/Jmol workflows.
Worked-case data: Plan a figure highlighting one chain’s helix region.

A. Primary sequence

B. Chemical bonds automatically

C. PDB identifier

D. Viewpoint

Answer: _____

4. (2 points) Quaternary structure requires…

Unit: Structural levels and denaturation.
Worked-case data: Heating disrupts activity but the amino-acid sequence remains intact. What changed?

A. No interfaces

B. Multiple polypeptide subunits

C. Exactly four atoms

D. Only one residue

Answer: _____

5. (2 points) An epitope is…

Unit: Viruses, entry, and immune recognition.
Worked-case data: An antibody binds a viral protein in an assay. Does that prove neutralization?

A. Every viral genome

B. A unit of salinity

C. A recognized molecular feature

D. The entire immune system

Answer: _____

6. (2 points) Color by structure shows…

Unit: Molecular viewing and reproducible JUDE/Jmol workflows.
Worked-case data: Plan a figure highlighting one chain’s helix region.

A. Measured free energy

B. Every atom’s charge

C. A diagnosis

D. A structural classification

Answer: _____

7. (2 points) Denaturation always cleaves peptide bonds?

Unit: Structural levels and denaturation.
Worked-case data: Heating disrupts activity but the amino-acid sequence remains intact. What changed?

A. Only at neutral pH

B. No

C. Always

D. Only for helices

Answer: _____

8. (2 points) Binding alone proves neutralization?

Unit: Viruses, entry, and immune recognition.
Worked-case data: An antibody binds a viral protein in an assay. Does that prove neutralization?

A. Only for any antibody

B. Only in a diagram

C. No

D. Always

Answer: _____

9. (2 points) The built-in lab executes full Jmol?

Unit: Molecular viewing and reproducible JUDE/Jmol workflows.
Worked-case data: Plan a figure highlighting one chain's helix region.

A. Yes

B. Only at maximum zoom

C. Only for HA

D. No

Answer: _____

10. (4 points) Compare motif and domain.

Unit: Structural levels and denaturation.
Worked-case data: Heating disrupts activity but the amino-acid sequence remains intact. What changed?

Answer: _____
_____
_____

11. (2 points) The assigned interval runs from residue 153 through 252, inclusive. How many numbered residue positions are included?

Answer: _____
_____
_____

12. (2 points) At the specified scale of 2 cm per residue, what backbone material length corresponds to 100 residues before folding?

Answer: _____
_____
_____

13. (2 points) State the assigned 2027 structure identifier, chain and residue interval, and distinguish these identifiers from a claim that all residues are experimentally resolved.

Answer: _____
_____
_____

14. (2 points) Explain the functional relevance of hemagglutinin binding to sialic-acid receptors in influenza entry.

Answer: _____
_____
_____

15. (2 points) Why does identifying an alpha helix require backbone geometry rather than simply finding a hydrophobic sequence stretch?

Answer: _____
_____
_____

16. (2 points) A synthetic structure worksheet places two oppositely charged side chains close together. Describe a possible stabilizing interaction and a qualification.

Answer: _____
_____
_____

17. (2 points) Explain why displaying a ligand-binding pocket without identifying chain and residue numbering can mislead a model reviewer.

Answer: _____
_____
_____

18. (2 points) What is the distinction between a displayed molecular contact and a proven functional interaction?

Answer: _____
_____
_____

19. (2 points) A model includes a decorative feature unrelated to protein structure or function. Explain why it does not improve its scientific explanation.

Answer: _____
_____
_____

20. (2 points) Explain why rotating a JUDE molecular view does not alter the primary sequence or chemical connectivity.

Answer: _____
_____
_____